

HealthConnect Clinic – Week 4 Project Summary

Problem Addressed

As a Data Analytics intern, my role in the HealthConnect Experience Lab is to understand the appointment dataset and identify how it can be used to investigate appointment attendance and missed appointment patterns. The goal is to establish a strong analytical foundation that can support future data-driven decisions aimed at reducing no-shows and improving the use of appointment slots.

Resources Used

The main resources reviewed for this stage of the project are the HealthConnect Appointment Dataset and the HealthConnect Clinic Knowledge Base. The appointment dataset contains fictional and anonymised information relating to patient demographics, appointment details, booking information, previous appointment history, previous no-shows, reminders, distance, waiting time, and appointment outcomes.

Key Observations

The HealthConnect business scenario identifies missed appointments as a key operational challenge. Understanding the overall appointment outcome distribution and investigating factors such as previous no-shows, reminders, waiting time, distance, patient characteristics, and appointment information may provide useful insights into attendance patterns.

Proposed Approach

My proposed approach is to first review the dataset structure and supporting Data Dictionary, conduct a data quality assessment, identify the most relevant variables, and define key business questions. Subsequent analysis will explore appointment outcomes and investigate patterns associated with missed appointments. The findings will later be communicated using appropriate KPIs, visualisations, dashboards, and business recommendations.

Key Considerations

Important considerations include data quality, missing information, duplicate records, inconsistent data formats, correct interpretation of appointment outcome categories, and the limitation that relationships identified in the data may indicate association rather than causation.

Proposed Week 5 Focus

In Week 5, I plan to continue with detailed exploratory data analysis, calculate the identified KPIs, investigate no-show patterns across important variables, and begin developing insights that can support recommendations for improving appointment attendance and patient engagement.